

SMOOTH QUARTIC HYPERSURFACES OF DIMENSION AT LEAST THREE IN PROJECTIVE SPACE ARE OKA

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ABSTRACT. We prove that every smooth hypersurface of degree four in complex projective space and of dimension at least three is an Oka manifold. The proof combines the geometry of lines on the hypersurface with holomorphic sprays associated with them and shows that their infinitesimal directions span every tangent space.

1. INTRODUCTION

Oka theory studies the flexibility and approximation properties of holomorphic maps on complex manifolds [6]. A central problem is to recognize this flexibility from concrete geometry. For projective manifolds, holomorphic sprays provide an effective criterion for the Oka property.

A holomorphic spray on a complex manifold X is a holomorphic map $s : \text{Tot}(E) \rightarrow X$ from the total space of a holomorphic vector bundle $E \rightarrow X$ such that $s(0_x) = x$ for every $x \in X$. A finite family of sprays is dominating when its infinitesimal directions span every tangent space. Gromov introduced ellipticity and proved that a complex manifold admitting a single dominating holomorphic spray is Oka [15, 0.6, p. 855]. Forstnerič and Prezelj subsequently established Oka principles for holomorphic fibre bundles and submersions with sprays [13, 14]. Forstnerič extended this implication from one dominating spray to a finite dominating family: every holomorphically subelliptic complex manifold is Oka [9, Theorem 1.1]. Conversely, Forstnerič and Lárusson proved that every smooth projective Oka manifold admits a single dominating holomorphic spray and is therefore holomorphically elliptic [10, Theorem 1.1]. Thus, for a smooth projective manifold, the Oka property is equivalent to holomorphic ellipticity, whereas a finite dominating family already suffices to prove the Oka property.

The remaining challenge is geometric: to construct globally defined holomorphic sprays whose infinitesimal directions span every tangent space.

Classical examples of projective Oka manifolds include projective complex homogeneous spaces, notably abelian varieties and rational homogeneous varieties [15, § 0.5]. More recently, Arzhantsev, Kaliman, and Zaidenberg considerably enlarged this class by proving that every complete uniformly rational variety is algebraically elliptic [1, Theorem 1.3 and Corollary 1.5]. For further examples and geometric criteria, see Forstnerič's surveys [7, 8].

Smooth hypersurfaces in projective space are far less homogeneous and form a more rigid testing ground. Smooth quadrics are homogeneous, while Kaliman and Zaidenberg proved that every smooth cubic hypersurface of dimension at least two is algebraically elliptic [17, Theorem 1.1]. Beyond projective space, Xie and Zhao constructed Oka hypersurfaces of multidegree $(2, \dots, 2)$ in $(\mathbb{P}^1)^{m+1}$, $m \geq 2$, including Oka K3 surfaces when $m = 2$ [23, Theorem B]. This raises the question whether smooth quartic hypersurfaces in projective space are Oka.

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Our main result settles this problem in dimension at least three; more precisely, it constructs a finite dominating family of line sprays.

Theorem 1.1. *Every smooth quartic hypersurface $X \subset \mathbb{P}^n$, $n \geq 4$, is an Oka manifold. More precisely, with $r = \binom{n+1}{2}$, there exist distinct lines $L_1, \dots, L_r \subset X$ whose Plücker bivectors form a basis of $\Lambda^2 \mathbb{C}^{n+1}$ and whose associated sprays*

$$s_{L_j} : \text{Tot}(\mathcal{O}_X(-2)) \longrightarrow X$$

form a dominating family.

The dominating family in the theorem makes X holomorphically subelliptic by definition. Forstnerič's subelliptic Oka principle then gives the Oka property [9, Theorem 1.1]; since X is projective, the theorem of Forstnerič and Lárusson further implies that X is holomorphically elliptic [10, Theorem 1.1].

The dimension range. The lower bound $\dim X \geq 3$ reflects a limitation of the line-based method: a general smooth quartic surface contains no line. The Oka problem for smooth quartic surfaces remains open in general, although the surface case is not entirely unknown. Xie and Zhao proved that the Fermat quartic is an Oka Kummer surface [23, Corollary 4.3] and conjectured that every K3 surface is Oka [23, Conjecture], which would settle the smooth quartic-surface case. Every line that does occur on a smooth quartic surface nevertheless determines an individual spray by the present construction; see Remark 9.2. For background on holomorphic flexibility of compact complex surfaces, see [12].

Proof strategy. The proof separates into two parts: an analytic construction of a spray from each line, and an algebraic argument showing that the resulting infinitesimal directions span every tangent space.

The three geometric stages of the construction are summarized in Figure 1.

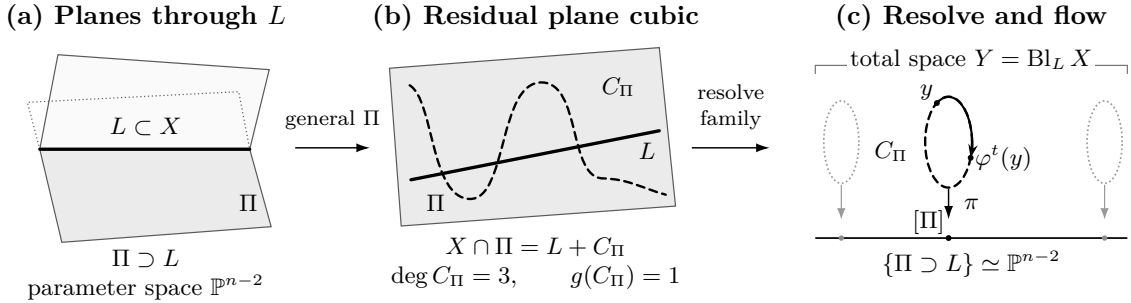


FIGURE 1. The line-projection construction. Planes $\Pi \supset L$ are parametrized by $\mathbb{P}^{n-2}$, and a general such plane cuts X in L and a smooth residual cubic C_Π . After blowing up L , projection resolves to $\pi : Y \rightarrow \mathbb{P}^{n-2}$, and the determinant direction integrates to the fibrewise flow φ^t . The picture is schematic; special fibres may be singular or higher-dimensional.

The guiding idea of the first part was inspired by Xie's construction for cubic complements [22, Sections 3–5], which in turn builds on the residual-conic geometry used by Kaliman and Zaidenberg for smooth cubic hypersurfaces [17, Proposition 3.7 and Remark 3.8]. The shared geometric mechanism is projection from a line. Planes through the line cut out residual curves; after resolving the projection, fibrewise flows along these curves produce holomorphic sprays.

The two implementations address different globalization problems. In Xie's noncompact setting, the main issue is to glue the Hamiltonian flows while keeping every trajectory away from

the deleted boundary. In the present compact setting, the main issue is to rescale the complete flow along the residual cubics so that it fixes the exceptional divisor and descends through the blowdown. The line sprays constructed here are holomorphic and generally nonalgebraic, in contrast to the algebraic sprays used for smooth cubics; this leaves open the possibility of a different algebraic spray on a quartic, as discussed in [Remark 9.3](#).

Construction of the line spray. Fix a line $L \subset X$, and let

$$\beta : Y = \mathrm{Bl}_L X \longrightarrow X$$

be the blowup morphism, with exceptional divisor D . Projection from L resolves to a proper map

$$\pi : Y \longrightarrow \mathbb{P}^{n-2},$$

whose general fibre is the residual plane cubic. Since the general fibre of π is one-dimensional, the determinant of $d\pi$ singles out a canonical tangent direction along the fibres. Properness makes the resulting holomorphic flow complete. The quartic identity

$$K_Y^{-1} \otimes \pi^* K_{\mathbb{P}^{n-2}} \cong \pi^* \mathcal{O}_{\mathbb{P}^{n-2}}(-2) \cong \beta^* \mathcal{O}_X(-2) \otimes \mathcal{O}_Y(2D)$$

provides the rescaling needed for this flow to fix D and descend through β to a spray on $\mathcal{O}_X(-2)$.

Domination. The second part of the proof is algebraic. Let $Z = F_1(X)$ denote the Fano scheme of lines on X . Its closed points parametrize the geometric lines contained in X , while its scheme structure records their infinitesimal deformations. Write $X = \{F = 0\} \subset \mathbb{P}(V)$, let $x = [v] \in X$, and denote by $dF_v \in V^*$ the differential of F at v . If $L = \mathbb{P}(\mathrm{Span}(a, b)) \subset X$, then the infinitesimal direction contributed by the line spray at $x = [v]$ is represented by

$$[dF_v(b)a - dF_v(a)b] \in \ker(dF_v)/\mathbb{C}v \cong T_x X.$$

Contraction with dF_v defines a surjective linear map

$$c_{dF_v} : \bigwedge^2 V \longrightarrow \ker(dF_v), \quad a \wedge b \longmapsto dF_v(b)a - dF_v(a)b.$$

Hence domination follows once the Plücker bivectors of the lines on X are shown to span $\bigwedge^2 V$.

For $n \geq 5$, Beheshti's dimension theorem and Larson's bound on the completely unbalanced locus imply that Z is reduced [[2](#), Theorem 4.2], [[19](#), Theorem 7.1]. A Koszul–Borel–Weil–Bott calculation then shows that Z is not contained in any Plücker hyperplane.

When $n = 4$, the Fano scheme Z may be nonreduced, so Plücker nondegeneracy of the scheme does not immediately imply spanning by its closed points. We prove that its nilradical is square-zero and then use a quadratic Koszul–Borel–Weil–Bott vanishing to bridge this gap.

Organization of the paper. Section [2](#) collects the background on sprays and the Grassmannian. Sections [3–5](#) construct the line sprays and compute their infinitesimal directions. Sections [6–8](#) establish the required Plücker spanning, with the quartic-threefold case treated separately. Section [9](#) completes the proof and discusses the scope of the method.

2. PRELIMINARIES

This section fixes the conventions on sprays and on the Grassmannian used throughout. Whenever holomorphic maps, holomorphic vector bundles, or sprays are involved, we understand a complex algebraic variety through its associated analytic space.

On projective space and on its subschemes, $\mathcal{O}_M(1)$ denotes the hyperplane line bundle; on a Grassmannian, it denotes the Plücker line bundle. For $k \in \mathbb{Z}$, set

$$\mathcal{O}_M(k) = \mathcal{O}_M(1)^{\otimes k}, \quad \mathcal{F}(k) = \mathcal{F} \otimes \mathcal{O}_M(k),$$

where negative tensor powers denote powers of the dual line bundle. The ideal sheaf of a closed subscheme $Z \subset M$ is $\mathcal{I}_{Z/M}$, and Z_{red} is its underlying reduced subscheme.

2.1. Sprays and Oka manifolds. Let $E \rightarrow X$ be a rank- e holomorphic vector bundle. We write $p_E : \text{Tot}(E) \rightarrow X$ for its total-space projection and $x \mapsto 0_x$ for its zero section.

Definition 2.1. Let X be a complex manifold. A *holomorphic spray* on X is a holomorphic vector bundle $E \rightarrow X$ together with a holomorphic map $s : \text{Tot}(E) \rightarrow X$ satisfying $s(0_x) = x$ for every $x \in X$. Its vertical differential differentiates s in the parameter fibre $E_x = p_E^{-1}(x)$ with x held fixed:

$$D_{\text{vert}}s(0_x) : E_x \longrightarrow T_x X, \quad \xi \longmapsto \left. \frac{d}{dt} \right|_{t=0} s(t\xi).$$

A finite collection $(E_j, s_j)_{j=1}^r$ is *dominating* if

$$\sum_{j=1}^r D_{\text{vert}}s_j(0_x)(E_{j,x}) = T_x X \quad \text{for every } x \in X.$$

The manifold X is *holomorphically subelliptic* if it admits such a finite dominating collection, and *holomorphically elliptic* if it admits one dominating spray.

For background on these notions, see [11, Section 5].

Over a trivializing open set $U \subset X$, one has $\text{Tot}(E|_U) \cong U \times \mathbb{C}^e$, and the vertical differential is the ordinary derivative in the $\mathbb{C}^e$ -direction.

After analytification, every algebraic spray is a holomorphic spray; in particular, algebraic ellipticity implies holomorphic ellipticity. More precisely, on a smooth algebraic variety, an *algebraic spray* is a spray whose parameter bundle and spray map are algebraic, and the variety is *algebraically elliptic* if one such spray is dominating, with domination understood as in Definition 2.1.

We use the convex approximation characterization of the Oka property: a complex manifold X is *Oka* if, for every $m \geq 1$, every compact convex set $K \subset \mathbb{C}^m$, and every holomorphic map f from a neighbourhood of K to X , there exist holomorphic maps $f_j : \mathbb{C}^m \rightarrow X$ converging uniformly to f on K ; see [5].

Gromov established the implication from ellipticity to the Oka property, Forstnerič extended it to subellipticity, and Forstnerič and Lárusson proved the converse for projective manifolds.

Theorem 2.2 (Gromov). *Every holomorphically elliptic complex manifold is Oka* [15, 0.6, p. 855].

Theorem 2.3 (Forstnerič). *Every holomorphically subelliptic complex manifold is Oka* [9, Theorem 1.1].

Theorem 2.4 (Forstnerič–Lárusson). *Every smooth projective Oka manifold is holomorphically elliptic* [10, Theorem 1.1].

Consequently, for smooth projective manifolds, the Oka property, holomorphic ellipticity, and holomorphic subellipticity are equivalent. The proof of the main theorem uses Forstnerič's subelliptic Oka principle and the projective converse of Forstnerič–Lárusson; Gromov's theorem records the original elliptic case.

2.2. The Grassmannian and the Fano scheme of lines. Our convention is that $\mathbb{P}(E)$ parametrizes one-dimensional subspaces of the fibres of a vector bundle E ; in particular, $\mathbb{P}(V)$ is the space of one-dimensional subspaces of V . Let V be a complex vector space of dimension $N \geq 5$, and set $G = \text{Gr}(2, V)$. In our applications, $N = n + 1$ and $\mathbb{P}(V) = \mathbb{P}^n$. A point of G is a two-dimensional subspace $A \subset V$, which corresponds to the line $\mathbb{P}(A) \subset \mathbb{P}(V)$.

Let $\mathcal{S}$ be the tautological rank-two bundle on G , with fibre $\mathcal{S}_A = A$. The Plücker line bundle is

$$\mathcal{O}_G(1) = \det \mathcal{S}^*,$$

and its complete linear system defines the Plücker embedding

$$G \hookrightarrow \mathbb{P}(\bigwedge^2 V), \quad A = \text{Span}(a, b) \mapsto [a \wedge b].$$

Moreover, $\dim G = 2N - 4$. By the Borel–Weil–Bott theorem, with the above convention for the tautological bundle [21, Corollary 4.1.9],

$$H^0(G, \mathcal{O}_G(1)) \cong \bigwedge^2 V^*.$$

Consequently, the Plücker image is linearly nondegenerate: no hyperplane of $\mathbb{P}(\bigwedge^2 V)$ contains G .

Let $F \in \text{Sym}^4 V^*$ be a quartic and put $X = \{F = 0\} \subset \mathbb{P}(V)$. Restricting F to the tautological subspaces $A \subset V$ gives a section

$$\sigma_F \in H^0(G, \mathcal{E}), \quad (\sigma_F)_A = F|_A, \quad \mathcal{E} = \text{Sym}^4 \mathcal{S}^*.$$

The zero scheme of σ_F is the Fano scheme of lines $Z = F_1(X)$. In a local frame a, b of $\mathcal{S}$, its five components are the coefficients of

$$F(sa + tb) = c_0 s^4 + c_1 s^3 t + c_2 s^2 t^2 + c_3 s t^3 + c_4 t^4$$

and cut out Z scheme-theoretically. Under a change of local frame of $\mathcal{S}$, these five functions transform as the components of σ_F ; their common zero scheme is therefore independent of the chosen frame. For a line $L \subset X$ we write $[L] \in Z$ for the corresponding point. The closed points of Z represent the geometric lines contained in X . When it is necessary to distinguish them from the infinitesimal information carried by the scheme structure of Z , we call them the actual lines.

A section of a rank- r vector bundle on a smooth variety is *regular* if its local components form a regular sequence along its zero scheme; that zero scheme is then a local complete intersection of codimension r , though it need not be smooth or reduced. This condition will be used in Section 7: the Koszul complex of a regular section resolves the ideal sheaf of its zero scheme.

3. DETERMINANT DIRECTIONS AND COMPLETE VERTICAL FLOWS

This section develops the analytic mechanism behind projection from a line. For a map of relative dimension one, the determinant of its differential produces a canonical vertical direction. When the map is proper and the parameter line bundle is pulled back from the base, this direction integrates for all complex times.

Let Y be a complex manifold of dimension m . Contraction of a vector with a holomorphic top form gives the canonical bundle isomorphism

$$TY \otimes K_Y \xrightarrow{\cong} \bigwedge^{m-1} T^*Y, \quad v \otimes \omega \mapsto \iota_v \omega. \quad (1)$$

Lemma 3.1. *Let $\pi : Y \rightarrow B$ be a holomorphic map between smooth complex manifolds with $\dim Y = m$ and $\dim B = m - 1$. Set*

$$\mathcal{T}_{Y/B} = \ker(d\pi : TY \rightarrow \pi^*TB),$$

viewed as a coherent subsheaf of TY . There is a canonical holomorphic morphism

$$\mu_\pi : K_Y^{-1} \otimes \pi^*K_B \rightarrow \mathcal{T}_{Y/B} \subset TY. \quad (2)$$

After composing μ_π with the inclusion $\mathcal{T}_{Y/B} \subset TY$, its value at a point $y \in Y$ is nonzero exactly when $d\pi_y$ has rank $m - 1$, and then its image in $T_y Y$ is the one-dimensional space $\ker d\pi_y$.

Proof. Taking $\wedge^{m-1}$ of the dual differential, tensoring with K_Y^{-1} , and using (1) produces (2). Choose local coordinates $b_1, \dots, b_{m-1}$ on B and a local volume form ω on Y . In these trivializations, the morphism μ_π is represented by the holomorphic vector field W characterized by

$$\iota_W \omega = d(b_1 \circ \pi) \wedge \cdots \wedge d(b_{m-1} \circ \pi).$$

Wedging with each $d(b_j \circ \pi)$ and using $\alpha \wedge \iota_W \omega = \alpha(W)\omega$ shows $d\pi(W) = 0$. The form on the right is nonzero exactly when $d\pi$ has rank $m - 1$; then W spans its one-dimensional kernel. $\square$

A vertical bundle morphism from a line bundle pulled back from the base integrates to a spray defined for all complex times, even when the fibres are singular or reducible. The following proposition is the rank-one case of the vertical integration construction of Xie and Zhao [23, Lemma 3.1], with properness stated explicitly to ensure the compactness of the fibres. Their holomorphic-volume-form construction [23, Section 3.3] also produces a vertical direction by contraction with the exterior power of the dual differential; in our application, the determinant direction supplies the vertical morphism.

Proposition 3.2. *Let $\pi : Y \rightarrow B$ be a proper holomorphic map between complex manifolds, let $M \rightarrow B$ be a holomorphic line bundle, and let*

$$\nu : \pi^* M \longrightarrow TY$$

be a holomorphic bundle morphism satisfying $d\pi \circ \nu = 0$. Locally trivializing M identifies a vector $\xi \in M_{\pi(y)}$ with a complex time parameter. These local flows are independent of the trivialization and glue to a holomorphic map

$$\Phi : \text{Tot}(\pi^* M) \longrightarrow Y$$

with the following properties:

- (i) $\Phi(y, 0) = y$ for every $y \in Y$;
- (ii) $\pi(\Phi(y, \xi)) = \pi(y)$ for every $\xi \in M_{\pi(y)}$;
- (iii) for every $y \in Y$, the vertical differential of Φ at 0_y , taken in the parameter fibre $M_{\pi(y)}$ with y fixed, is ν_y .

Thus Φ is a holomorphic spray on Y preserving the fibres of π .

For the reader's convenience, we include the proof.

Proof. Let e be a nowhere-vanishing frame of M over $U \subset B$. The morphism ν defines a vertical holomorphic vector field $W_e(y) = \nu_y(e(\pi(y)))$ on $\pi^{-1}(U)$; its local flow ψ_e^τ preserves each fibre of π .

Fix $U_0 \Subset U$. Properness makes $K = \pi^{-1}(\overline{U_0})$ compact. The holomorphic ODE theorem and compactness give $r > 0$ and a neighbourhood Ω of K on which $(\tau, y) \mapsto \psi_e^\tau(y)$ is holomorphic for $|\tau| < r$. Since W_e is vertical, every trajectory starting in K remains in the same fibre and hence in K for as long as it is defined. Thus the uniform short-time flow can be iterated without leaving its domain. For any $R > 0$, choose k with $R/k < r$ and define

$$(t, y) \longmapsto (\psi_e^{t/k})^{\circ k}(y), \quad |t| < R, \quad y \in \pi^{-1}(U_0).$$

Every intermediate point stays in K , so this map is well defined and jointly holomorphic. The local flow law, applied after passage to a common multiple of k , makes the maps independent of k and compatible as R and U_0 vary. They yield an entire, jointly holomorphic flow $\varphi_e^t(y)$ on $\mathbb{C} \times \pi^{-1}(U)$, with no smoothness assumption on the fibres.

On an overlap, write $e' = he$. Then $W_{e'} = (h \circ \pi)W_e$, and $h \circ \pi$ is constant along trajectories. If $\xi = t'e' = te \in M_{\pi(y)}$, then $t = t'h(\pi(y))$, and hence

$$\varphi_{e'}^{t'}(y) = \varphi_e^{t'h(\pi(y))}(y) = \varphi_e^t(y).$$

Thus the local flows define the same map for the same vector ξ and glue to the required $\Phi : \text{Tot}(\pi^*M) \rightarrow Y$. Its value and vertical differential at zero follow from the flow equation. $\square$

Remark 3.3. No regularity of the fibres of π is used. At a critical point of π , the determinant direction from [Lemma 3.1](#) vanishes. Completeness comes from compactness of the containing fibre in the smooth ambient manifold Y . The pullback form of the parameter bundle makes the gluing argument work: its transition functions are constant along the vertical trajectories.

4. THE SPRAY ASSOCIATED WITH A LINE ON A QUARTIC

We now carry out the construction depicted in [Figure 1](#). Let $X = \{F = 0\} \subset \mathbb{P}^n$ be a smooth quartic hypersurface with $n \geq 4$, and fix a line $L \subset X$. Blowing up L resolves projection from L . Write

$$\beta : Y = \text{Bl}_L X \longrightarrow X$$

for the blowup along L , with exceptional divisor $D \subset Y$. Let π denote the resulting morphism to the space of planes through L , and let H be the pullback to Y of the hyperplane Cartier divisor class on X , so that

$$\mathcal{O}_Y(H) \cong \beta^* \mathcal{O}_X(1).$$

Lemma 4.1. *Projection from L extends to a morphism*

$$\pi : Y \longrightarrow B = \mathbb{P}^{n-2}.$$

This morphism is proper, Y is smooth, and

$$\pi^* \mathcal{O}_B(1) \cong \mathcal{O}_Y(H - D), \tag{3}$$

$$K_Y \cong \mathcal{O}_Y((3 - n)H + (n - 3)D), \tag{4}$$

$$K_Y^{-1} \otimes \pi^* K_B \cong \pi^* \mathcal{O}_B(-2). \tag{5}$$

Proof. Choose coordinates with $L = \{x_2 = \dots = x_n = 0\}$. The forms $x_2, \dots, x_n$ generate $\mathcal{I}_{L/X} \otimes \mathcal{O}_X(1)$. On the blowup, $\mathcal{I}_{L/X} \mathcal{O}_Y = \mathcal{O}_Y(-D)$ [[20, Tag 02OS](#)]. Let σ_D be the canonical section of $\mathcal{O}_Y(D)$ with zero divisor D . The pullbacks $\beta^* x_2, \dots, \beta^* x_n$ vanish along D . Dividing them by σ_D gives sections

$$\tilde{x}_i = \beta^* x_i / \sigma_D \in H^0(Y, \mathcal{O}_Y(H - D)).$$

By the defining property of the blowup, these sections have no common zero. They define $\pi : Y \rightarrow \mathbb{P}^{n-2}$, extending projection from L , and give [\(3\)](#).

Since both X and L are smooth, the blowup $Y = \text{Bl}_L X$ is smooth [[20, Tag 0FUT](#)]. Moreover, a blowup is projective over the original variety [[20, Tag 02NS](#)]. Hence Y is projective and π is proper.

Adjunction gives $K_X = \mathcal{O}_X(3 - n)$. Since L has codimension $n - 2$ in X , the discrepancy of the blowup is

$$\text{codim}_X L - 1 = n - 3.$$

The canonical-bundle formula [[18, § 4, Example 27](#)] gives

$$K_Y = \beta^* K_X \otimes \mathcal{O}_Y((n - 3)D),$$

which is [\(4\)](#). Finally, $K_B = \mathcal{O}_B(-(n - 1))$ and [\(3\)](#) yield

$$K_Y^{-1} \otimes \pi^* K_B \cong \mathcal{O}_Y((n - 3)(H - D) - (n - 1)(H - D)) \cong \pi^* \mathcal{O}_B(-2).$$

□

The fibres of π have a simple description, shown schematically in panels (b) and (c) of [Figure 1](#). For a plane P_b not contained in X , the fibre is the cubic residual to L in $P_b \cap X$. If $P_b \subset X$, the fibre jumps to the whole plane P_b .

Lemma 4.2. *For $b \in B$, let $P_b \subset \mathbb{P}^n$ be the plane through L corresponding to b under projection from L . If $P_b \not\subset X$, there is a unique plane cubic $C_b \subset P_b$ such that*

$$P_b \cap X = L + C_b \quad (6)$$

as effective Cartier divisors on P_b . The restriction of β induces an isomorphism of schemes

$$\beta_b : \pi^{-1}(b) \xrightarrow{\cong} C_b. \quad (7)$$

If $P_b \subset X$, then β instead identifies $\pi^{-1}(b)$ with P_b . For general $b \in B$ the plane P_b is not contained in X , the fibre $\pi^{-1}(b) \cong C_b$ is a smooth plane cubic, and π is smooth along this fibre.

Proof. Fix $b \in B$ and choose $\tau_b \in H^0(P_b, \mathcal{O}_{P_b}(1))$ with $V(\tau_b) = L$. If $P_b \not\subset X$, divisibility by τ_b gives

$$F|_{P_b} = \tau_b G_b, \quad 0 \neq G_b \in H^0(P_b, \mathcal{O}_{P_b}(3)).$$

Thus $C_b := V(G_b)$ is independent of the choice of τ_b and is the unique plane cubic satisfying (6).

Consider the ambient blowup

$$\tilde{\beta} : \tilde{P} = \text{Bl}_L \mathbb{P}^n \longrightarrow \mathbb{P}^n,$$

and denote its exceptional divisor by $\tilde{D}$. Projection from L extends to a morphism

$$\rho : \tilde{P} \longrightarrow B.$$

We describe its fibres explicitly, since the same model identifies the fibres of π . Give B homogeneous coordinates $[y_2 : \cdots : y_n]$. The blowup $\tilde{P}$ is the closure of the graph of

$$[x_0 : \cdots : x_n] \dashrightarrow [x_2 : \cdots : x_n],$$

and hence is cut out in $\mathbb{P}^n \times B$ by

$$x_i y_j = x_j y_i \quad (2 \leq i, j \leq n).$$

For a fixed $b = [y_2 : \cdots : y_n]$, these equations say that $(x_2, \dots, x_n) = \lambda(y_2, \dots, y_n)$. Thus

$$\rho^{-1}(b) \cong \mathbb{P}^2, \quad [x_0 : x_1 : \lambda] \longmapsto [x_0 : x_1 : \lambda y_2 : \cdots : \lambda y_n],$$

and $\tilde{\beta}$ identifies this fibre with the plane P_b through L . The same coordinates, on each chart $y_i \neq 0$ of B , give a local trivialization of ρ as a $\mathbb{P}^2$ -bundle. Moreover, $\tilde{D} \cap \rho^{-1}(b)$ is the line $\{\lambda = 0\}$, identified with $L \subset P_b$. By the strict-transform property of blowups [[20, Tag 080E](#)], $Y = \text{Bl}_L X$ is the strict transform of X in $\tilde{P}$. At every point $p \in L$, a local equation f of X satisfies

$$f \in \mathcal{I}_{L/\mathbb{P}^n, p} \setminus \mathcal{I}_{L/\mathbb{P}^n, p}^2.$$

Indeed, membership in the square would make df_p vanish in the normal directions to L , while $f|_L = 0$ already makes it vanish in the tangent directions; this would make p singular on X . Thus the pullback of f has exactly one exceptional factor. Writing $\sigma_{\tilde{D}}$ for the canonical section of the exceptional divisor, we have

$$\tilde{\beta}^* F = \sigma_{\tilde{D}} \tilde{F}, \quad Y = V(\tilde{F}), \quad \pi = \rho|_Y.$$

Under $\rho^{-1}(b) \cong P_b$, the divisor $\tilde{D}|_{\rho^{-1}(b)}$ is L ; hence $\sigma_{\tilde{D}}|_{\rho^{-1}(b)} = c\tau_b$ for some $c \neq 0$. If $P_b \not\subset X$, restriction of the displayed identity gives $\tau_b G_b = c\tau_b \tilde{F}|_{\rho^{-1}(b)}$. The form τ_b is a non-zero-divisor in the homogeneous coordinate ring of the plane P_b , so it can be cancelled. The resulting equality is

scheme-theoretic: the fibre $V(\tilde{F}|_{\rho^{-1}(b)})$ is C_b . If $P_b \subset X$, the same cancellation gives $\tilde{F}|_{\rho^{-1}(b)} = 0$, so the fibre is all of P_b .

Every fibre is nonempty, so π is dominant. Both Y and B are smooth varieties over the characteristic-zero field $\mathbb{C}$. Generic smoothness therefore gives a dense open subset $B^\circ \subset B$ over which π is smooth. Since

$$\dim Y - \dim B = 1,$$

the fibres over B° are smooth residual plane cubics. $\square$

The equality in (6) is a sum of effective Cartier divisors, not in general a scheme-theoretic union. Special fibres cause no difficulty for the flow construction. A residual cubic may be singular, reducible, or nonreduced, and a plane contained in X gives a two-dimensional fibre. At every point where $d\pi$ has rank less than $n - 2$, the determinant direction vanishes by Lemma 3.1. Thus the construction requires properness of π , but neither smoothness nor equidimensionality of its fibres.

Fix an isomorphism in (5). This choice rescales the resulting vertical bundle morphism by a nonzero constant. Hence the spray itself depends on the choice, but its infinitesimal direction does not. The determinant morphism of Lemma 3.1 then becomes a holomorphic bundle morphism

$$\nu : \pi^* \mathcal{O}_B(-2) \longrightarrow TY, \quad (8)$$

and Proposition 3.2 gives a holomorphic map on the entire time bundle,

$$\Phi : \text{Tot}(\pi^* \mathcal{O}_B(-2)) \longrightarrow Y. \quad (9)$$

It fixes the zero section and preserves π .

We shall descend a reparametrization of (9). We record the standard descent statement in the form needed here; for a related parameter-dependent version, see [22, Lemma 5.2].

Lemma 4.3. *Let $q : Z \rightarrow W$ be a proper surjective bimeromorphic holomorphic map, where Z and W are complex manifolds. Let T be a complex manifold and let $A : Z \rightarrow T$ be holomorphic. If A is constant on every fibre of q , then there is a unique holomorphic map $a : W \rightarrow T$ satisfying $A = a \circ q$.*

Proof. Fibrewise constancy defines a set map $a : W \rightarrow T$, which is continuous because a proper surjection is a quotient map. Since W is normal and q is a proper modification, the analytic Riemann extension theorem gives

$$q_* \mathcal{O}_Z = \mathcal{O}_W.$$

For the same descent mechanism, including parameters, see [22, Lemma 5.2]. Near any $w \in W$, choose a coordinate chart of T containing $a(w)$ and shrink around w so that its image remains in that chart. Each coordinate of A then descends through $q_* \mathcal{O}_Z = \mathcal{O}_W$ to a holomorphic function. These coordinates define the required holomorphic factorization. Surjectivity of q gives uniqueness. $\square$

By (3),

$$\pi^* \mathcal{O}_B(-2) \cong \beta^* \mathcal{O}_X(-2) \otimes \mathcal{O}_Y(2D).$$

Fix this isomorphism as well. Recall that σ_D is the canonical section of $\mathcal{O}_Y(D)$ with zero divisor D . The exponent two is forced by the exceptional twist $2D$ in the displayed isomorphism. Multiplication by σ_D^2 therefore converts a parameter in $\beta^* \mathcal{O}_X(-2)$ into a time parameter for the complete vertical flow and gives a holomorphic line-bundle morphism

$$j : \beta^* \mathcal{O}_X(-2) \longrightarrow \pi^* \mathcal{O}_B(-2), \quad j_y(\xi) = \xi \otimes \sigma_D(y)^2. \quad (10)$$

It is an isomorphism away from D and is identically zero over D .

Proposition 4.4. *Every line $L \subset X$ determines a holomorphic spray*

$$s_L : \text{Tot}(\mathcal{O}_X(-2)) \longrightarrow X.$$

For every $x \in L$,

$$s_L(\mathcal{O}_X(-2)_x) = \{x\}.$$

At $x \in X \setminus L$, if projection from L is a submersion on X at x , then the image of $D_{\text{vert}} s_L(0_x)$ is the tangent line at x to the projection fibre through x .

Proof. On $\text{Tot}(\beta^* \mathcal{O}_X(-2))$, set

$$S(y, \xi) = \beta(\Phi(y, j_y(\xi))). \quad (11)$$

Consider the natural map

$$q : \text{Tot}(\beta^* \mathcal{O}_X(-2)) \longrightarrow \text{Tot}(\mathcal{O}_X(-2)), \quad q(y, \xi) = (\beta(y), \xi).$$

It is the base change of β by the vector-bundle projection $\text{Tot}(\mathcal{O}_X(-2)) \rightarrow X$. Explicitly,

$$q^{-1}(x, \xi) = \{(y, \xi) : y \in \beta^{-1}(x)\},$$

so every fibre of q is naturally identified with a fibre of β . Hence q is proper, surjective, and bimeromorphic.

The map S is constant on every fibre of q . If $x \notin L$, then $\beta^{-1}(x)$ consists of one point. If $x \in L$ and $y \in \beta^{-1}(x)$, then $y \in D$, so (10) gives $j_y(\xi) = 0$ for every $\xi \in \mathcal{O}_X(-2)_x$; since Φ fixes its zero section,

$$S(y, \xi) = \beta(\Phi(y, 0)) = \beta(y) = x.$$

Thus Lemma 4.3 gives a unique holomorphic map $s_L : \text{Tot}(\mathcal{O}_X(-2)) \rightarrow X$ with $S = s_L \circ q$. For $\xi = 0$, the same formula shows $s_L(0_x) = x$, so s_L is a spray, and it is constant on the whole parameter fibre over $x \in L$.

Let $x \in X \setminus L$ and put $y = \beta^{-1}(x)$. The maps $d\beta_y$ and j_y are isomorphisms in the relevant directions, and differentiating (11) along the parameter fibre at zero gives, with the line-bundle isomorphisms chosen above,

$$D_{\text{vert}} s_L(0_x) = d\beta_y \circ \nu_y \circ j_y : \mathcal{O}_X(-2)_x \longrightarrow T_x X.$$

Changing either of the chosen line-bundle isomorphisms multiplies this differential by a nonzero scalar. Its image in $T_x X$, which is the only datum used in the domination argument, is therefore intrinsic. At a submersion point of π the determinant direction is nonzero and spans $\ker d\pi_y$, by Lemma 3.1; its image under $d\beta_y$ is exactly the tangent line to the projection fibre through x . $\square$

Remark 4.5. The descended spray need not satisfy a group law and should not be regarded as the flow of a local vector-field representative of $\nu \circ j$. Indeed, the factor σ_D^2 in (10) rescales only the *initial time* parameter of the already complete flow Φ ; it is not asserted to remain constant along its trajectories.

We have now constructed one spray from every line $L \subset X$. The next section identifies its infinitesimal direction with the contraction of the Plücker bivector of L , reducing domination to a linear spanning problem in $\bigwedge^2 V$.

5. INFINITESIMAL DIRECTIONS AND PLÜCKER SPANNING

We compute the infinitesimal direction supplied by the spray of a line $L \subset X$ and identify it with a contraction of the Plücker bivector of L .

The tangent space of $X = \{F = 0\} \subset \mathbb{P}(V)$ can be described via its homogeneous defining polynomial $F : V \rightarrow \mathbb{C}$. Fix $x = [v] \in X$ and choose a nonzero representative $v \in V$. The differential of F at v is the linear functional

$$g = dF_v \in V^*, \quad g(u) = \left. \frac{d}{dt} \right|_{t=0} F(v + tu).$$

Smoothness gives $g \neq 0$. A variation $v + tu$ preserves the equation $F = 0$ to first order precisely when $g(u) = 0$. Since F is homogeneous of degree four, Euler's identity gives $g(v) = 4F(v) = 0$, so $\mathbb{C}v \subset \ker g$; a displacement proportional to v therefore gives the zero projective tangent vector. Intrinsically, $T_x X = \text{Hom}(\mathbb{C}v, \ker g/\mathbb{C}v)$. After choosing the representative v , evaluation at v gives

$$T_x X \xrightarrow{\cong} \ker g/\mathbb{C}v. \quad (12)$$

The class of u corresponds to the tangent vector represented by $t \mapsto [v + tu]$ in projective space.

Let $L \subset X$ be a line, and choose a basis a, b of the two-dimensional vector subspace $A \subset V$ with $L = \mathbb{P}(A)$. Set

$$w_L(v) = g(b)a - g(a)b \in \ker g. \quad (13)$$

Changing the ordered basis (a, b) multiplies this vector by the determinant of the basis change. If v is replaced by cv , homogeneity gives

$$dF_{cv} = c^3 dF_v, \quad w_L(cv) = c^3 w_L(v).$$

Meanwhile, evaluation at cv instead of v changes the representative of a fixed tangent vector by a factor c . Hence the tangent vector represented by $w_L(cv)$ is c^2 times the one represented by $w_L(v)$. In particular, its vanishing and the line it spans in $T_x X$ are independent of the chosen representative.

Proposition 5.1. *For every $x = [v] \in X$, the image of the vertical differential $D_{\text{vert}} s_L(0_x)$ (see Definition 2.1) is the vector subspace of (12) spanned by the class of $w_L(v)$. It is zero if that class vanishes and is one-dimensional otherwise.*

Proof. Take $x \notin L$ first, so $v \notin A$. The ambient projection from L is induced by $V \rightarrow V/A$, and its differential at $[v]$ has kernel

$$(A + \mathbb{C}v)/\mathbb{C}v \subset T_{[v]} \mathbb{P}(V).$$

Intersecting with $T_x X = \ker g/\mathbb{C}v$ amounts to imposing the equation $g = 0$ on A .

Case $g|_A \neq 0$. Then $\ker g|_A$ is one-dimensional, generated by $g(b)a - g(a)b$. The projection is therefore a submersion on X at x , and Proposition 4.4 identifies the spray direction with the class of (13).

Case $g|_A = 0$. Here

$$(A + \mathbb{C}v)/\mathbb{C}v \subset T_x X$$

has dimension two. Thus the kernel of the restricted differential of projection from L is larger than its generic one-dimensional kernel, so the rank drops. The determinant direction therefore vanishes by Lemma 3.1, and $w_L(v) = 0$ as well.

Case $x \in L$. Then $v \in A$. Since $F|_A$ is identically zero, its differential at v annihilates A , and therefore $w_L(v) = 0$. The spray is constant on the parameter fibre over such an x by Proposition 4.4, so its vertical differential is also zero. $\square$

For $\phi \in V^*$, define the contraction map

$$c_\phi : \bigwedge^2 V \longrightarrow \ker \phi, \quad c_\phi(a \wedge b) = \phi(b)a - \phi(a)b, \quad (14)$$

so that $c_g(a \wedge b) = g(b)a - g(a)b$ for $g = dF_v$ as above.

Lemma 5.2. *The map c_ϕ is surjective for every $\phi \neq 0$.*

Proof. Pick $u \in V$ with $\phi(u) = 1$. For every $w \in \ker \phi$,

$$c_\phi(w \wedge u) = \phi(u)w - \phi(w)u = w.$$

Since w was arbitrary, $c_\phi : \bigwedge^2 V \rightarrow \ker \phi$ is surjective. $\square$

Proposition 5.3. *Let V be a complex vector space with $\dim V \geq 5$, and let $X \subset \mathbb{P}(V)$ be a smooth quartic hypersurface. Suppose that X contains lines $L_1, \dots, L_r$ whose Plücker bivectors span $\bigwedge^2 V$. Then the corresponding sprays $s_{L_1}, \dots, s_{L_r}$ form a dominating family on X . Consequently X is Oka and holomorphically elliptic.*

Proof. From the assumed spanning set of Plücker bivectors, choose a basis of $\bigwedge^2 V$. It consists of

$$\dim \bigwedge^2 V = \binom{\dim V}{2}$$

bivectors associated with distinct lines on X . It is enough to use the sprays associated with these lines.

Fix $x = [v] \in X$ and set $g = dF_v$. By Lemma 5.2, the contractions of the spanning Plücker bivectors span $\ker g$. Passing to the quotient by $\mathbb{C}v$, their classes span $T_x X$. By Proposition 5.1 the image of each vertical differential is spanned by the corresponding class, a vanishing contraction contributing the zero vector; so the sprays dominate at x . Since x was arbitrary, the family is dominating on all of X , so X is holomorphically subelliptic. Forstnerič's theorem implies that X is Oka. Since X is projective, the theorem of Forstnerič–Lárusson then implies that X is holomorphically elliptic. $\square$

Although the spray s_L has zero vertical differential along its defining line L , the sprays associated with all the chosen lines still dominate at every $x \in X$, by the Plücker spanning argument.

6. FANO SCHEMES OF LINES ON SMOOTH QUARTICS

Here $F_1(X)$ denotes the closed subscheme of the Hilbert scheme of X parametrizing subschemes with Hilbert polynomial $t + 1$, equivalently the lines contained in X . Via the Plücker embedding, it is the zero scheme $Z \subset G = \text{Gr}(2, n + 1)$ described in Section 2. Its closed points, or equivalently the closed points of Z_{red} , are the actual geometric lines on X ; the nilpotent structure of Z records infinitesimal information and must be kept separate from this point set.

For every smooth quartic hypersurface $X \subset \mathbb{P}^n$, $n \geq 4$, the scheme $Z = F_1(X)$ is an expected-dimensional local complete intersection; when $n \geq 5$, it is moreover reduced.

Proposition 6.1. *Let $X \subset \mathbb{P}^n$, $n \geq 4$, be a smooth quartic hypersurface. Then $Z = F_1(X)$ is a nonempty, pure-dimensional regular zero locus of $\mathcal{E} = \text{Sym}^4 \mathcal{S}^*$, of dimension $2n - 7$. In particular, Z is a local complete intersection and is Cohen–Macaulay. If $n \geq 5$, then Z is reduced.*

Here “regular zero locus” means that the defining section σ_F is regular, or equivalently that its five local components form a regular sequence along Z . It does not mean that Z is smooth.

Proof. Beheshti proved that, over an algebraically closed field of characteristic zero, the Fano scheme of lines on every smooth hypersurface of degree $d \leq 6$ in $\mathbb{P}^n$ has the expected dimension $2n - d - 3$ whenever $d \leq n$ [2, Theorem 4.2]. Applying this with $d = 4$ gives

$$\dim Z = 2n - 7 \quad (15)$$

for $Z = F_1(X)$. Since $2n - 7 \geq 1$ for $n \geq 4$, this equality also implies that Z is nonempty.

For quartic threefolds, the expected-dimension result goes back to Collino [3]; Beheshti also records Collino's argument as the antecedent for the degree-four case.

The scheme Z is locally defined in the smooth Grassmannian G by the five components of the section σ_F , so every irreducible component has codimension at most five. The dimension formula above therefore forces every component to have codimension exactly five. Thus the five local generators have height five in the regular local rings of G and form a regular sequence. Hence Z is a pure-dimensional local complete intersection and, in particular, Cohen–Macaulay.

For a line $L \subset X$, let $N_{L/X} = TX|_L/TL$ be its normal bundle. The inclusions $L \subset X \subset \mathbb{P}^n$ give the normal bundle sequence

$$0 \longrightarrow N_{L/X} \longrightarrow N_{L/\mathbb{P}^n} \longrightarrow N_{X/\mathbb{P}^n}|_L \longrightarrow 0.$$

Since

$$N_{L/\mathbb{P}^n} \cong \mathcal{O}_L(1)^{\oplus(n-1)}, \quad N_{X/\mathbb{P}^n}|_L \cong \mathcal{O}_L(4),$$

this becomes

$$0 \longrightarrow N_{L/X} \longrightarrow \mathcal{O}_L(1)^{\oplus(n-1)} \longrightarrow \mathcal{O}_L(4) \longrightarrow 0. \quad (16)$$

The right-hand map is induced by the differential of the quartic equation F . Write

$$N_{L/X} \cong \bigoplus_{i=1}^{n-2} \mathcal{O}_L(a_i), \quad a_1 \leq \cdots \leq a_{n-2}.$$

The inclusion $N_{L/X} \hookrightarrow \mathcal{O}_L(1)^{\oplus(n-1)}$ restricts to a nonzero map $\mathcal{O}_L(a_i) \rightarrow \mathcal{O}_L(1)^{\oplus(n-1)}$ on every direct summand. Such a map can exist only when $a_i \leq 1$. For a coherent sheaf $\mathcal{F}$ on the projective line L , write $h^i(L, \mathcal{F}) = \dim_{\mathbb{C}} H^i(L, \mathcal{F})$ and $\chi(L, \mathcal{F}) = h^0(L, \mathcal{F}) - h^1(L, \mathcal{F})$. We suppress L in $\chi(N_{L/X})$. Sequence (16) gives

$$a_i \leq 1, \quad \sum_{i=1}^{n-2} a_i = n - 5, \quad \chi(N_{L/X}) = 2n - 7. \quad (17)$$

By the standard deformation theory of the Hilbert scheme, first-order deformations of L inside X are parametrized by $H^0(L, N_{L/X})$. Hence

$$T_{[L]}Z \cong H^0(L, N_{L/X}).$$

Since Z is pure of local dimension $2n - 7$, the point $[L]$ is smooth precisely when $\dim T_{[L]}Z = 2n - 7$. Riemann–Roch on $L \cong \mathbb{P}^1$ gives

$$h^0(L, N_{L/X}) = \chi(N_{L/X}) + h^1(L, N_{L/X}) = 2n - 7 + h^1(L, N_{L/X}).$$

Thus $[L]$ is singular precisely when $H^1(L, N_{L/X}) \neq 0$. By the ordering of the a_i , this is equivalent to $a_1 \leq -2$. The deficit from the maximal possible value one is

$$\sum_{i=1}^{n-2} (1 - a_i) = (n - 2) - \sum_{i=1}^{n-2} a_i = (n - 2) - (n - 5) = 3,$$

so this happens precisely for the *completely unbalanced* splitting type

$$N_{L/X} \cong \mathcal{O}_L(-2) \oplus \mathcal{O}_L(1)^{\oplus(n-3)}. \quad (18)$$

For a smooth degree- d hypersurface $X \subset \mathbb{P}^n$, let

$$F'(X) = \{[L] \in F_1(X) : N_{L/X} \cong \mathcal{O}_L(2-d) \oplus \mathcal{O}_L(1)^{\oplus(n-3)}\}.$$

Larson proved [19, Theorem 7.1] that, for every $d \geq 3$,

$$\dim F'(X) \leq n-3.$$

For $d = 4$, this is precisely the locus with normal bundle (18), and hence

$$\dim\{[L] \in Z : N_{L/X} \text{ has type (18)}\} \leq n-3. \quad (19)$$

If $n \geq 5$, then

$$(2n-7) - (n-3) = n-4 > 0.$$

Thus the singular locus contains no generic point of a component of Z , and Z is generically smooth, hence generically reduced. Since Z is Cohen–Macaulay, it satisfies Serre’s condition (S_1) ; generic smoothness gives (R_0) . A Noetherian scheme satisfying (R_0) and (S_1) is reduced, so Z is reduced. $\square$

When $n = 4$, the bundle $N_{L/X}$ has rank two. Writing its splitting type as $a_1 \leq a_2$, the constraints in (17) become $a_1 + a_2 = -1$ and $0 \leq a_2 \leq 1$. The two possible normal bundles are therefore

$$N_{L/X} \cong \mathcal{O}_L \oplus \mathcal{O}_L(-1) \quad \text{or} \quad N_{L/X} \cong \mathcal{O}_L(1) \oplus \mathcal{O}_L(-2). \quad (20)$$

The second type can occur along an entire component, and the Fano scheme can then be generically nonreduced, so the reducedness argument above does not apply to quartic threefolds. More precisely, Lemma 8.1 will show that a component whose general line has the second type has generic length two.

For $n \geq 5$, the reducedness just proved allows linear Plücker nondegeneracy to be read directly from the actual lines. For $n = 4$, we instead combine a quadratic nondegeneracy statement with the generic multiplicity bound above.

7. KOSZUL RESOLUTIONS AND PLÜCKER NONDEGENERACY

We first prove that a regular zero scheme of $\text{Sym}^4 \mathcal{S}^*$ has no linear Plücker equation. This will settle the spanning problem once the scheme is reduced. On $\text{Gr}(2, 5)$ we then prove the stronger absence of quadratic equations; in Section 8, the square-zero nilradical of the quartic-threefold line scheme turns that statement into spanning by its actual lines.

By Proposition 6.1, the Fano scheme $F_1(X)$ is a regular zero scheme of $\text{Sym}^4 \mathcal{S}^*$. For the cohomological calculation, we work more generally with the zero scheme $Z \subset G$ of any regular section $\sigma \in H^0(G, \mathcal{E})$, where $\mathcal{E} = \text{Sym}^4 \mathcal{S}^*$. The Koszul complex of σ resolves its ideal sheaf $\mathcal{I}_Z = \mathcal{I}_{Z/G}$. After tensoring by $\mathcal{O}_G(1) = \det \mathcal{S}^*$ we obtain

$$0 \longrightarrow K_5 \longrightarrow K_4 \longrightarrow K_3 \longrightarrow K_2 \longrightarrow K_1 \longrightarrow \mathcal{I}_Z(1) \longrightarrow 0, \quad (21)$$

where

$$K_p = \bigwedge^p (\text{Sym}^4 \mathcal{S}) \otimes \det \mathcal{S}^*, \quad 1 \leq p \leq 5. \quad (22)$$

Thus K_p is the term in homological degree p , and each differential is induced by contraction with σ . The image of $K_1 \rightarrow \mathcal{O}_G(1)$ is $\mathcal{I}_Z(1)$, which accounts for the last term in (21).

It is enough to prove

$$H^{p-1}(G, K_p) = 0 \quad (1 \leq p \leq 5), \quad (23)$$

since these vanishings together with (21) imply $H^0(G, \mathcal{I}_Z(1)) = 0$. Indeed, the long exact cohomology sequences of the successive image sheaves in the Koszul complex link $H^0(G, \mathcal{I}_Z(1))$ to $H^4(G, K_5)$; at the p -th step, the vanishing of $H^{p-1}(G, K_p)$ makes the connecting map injective. The complete dimension-shifting argument is given in the proof of Proposition 7.4.

7.1. Exterior powers of the binary quartic representation.

Lemma 7.1. *The following $\mathrm{GL}_2(\mathbb{C})$ -equivariant identities induce decompositions for every rank-two complex vector bundle S :*

$$\bigwedge^1 \mathrm{Sym}^4 S \cong \mathrm{Sym}^4 S, \quad (24)$$

$$\bigwedge^2 \mathrm{Sym}^4 S \cong (\mathrm{Sym}^6 S \otimes \det S) \oplus (\mathrm{Sym}^2 S \otimes (\det S)^3), \quad (25)$$

$$\bigwedge^3 \mathrm{Sym}^4 S \cong (\mathrm{Sym}^6 S \otimes (\det S)^3) \oplus (\mathrm{Sym}^2 S \otimes (\det S)^5), \quad (26)$$

$$\bigwedge^4 \mathrm{Sym}^4 S \cong \mathrm{Sym}^4 S \otimes (\det S)^6, \quad (27)$$

$$\bigwedge^5 \mathrm{Sym}^4 S \cong (\det S)^{10}. \quad (28)$$

Proof. Let x, y be the two coordinate characters of the diagonal torus in $\mathrm{GL}_2(\mathbb{C})$, and write ch for a representation's character. The weights of Sym^4 are $x^4, x^3y, x^2y^2, xy^3, y^4$; summing their ten pairwise products gives

$$\mathrm{ch}\left(\bigwedge^2 \mathrm{Sym}^4\right) = xy \, \mathrm{ch}(\mathrm{Sym}^6) + x^3y^3 \, \mathrm{ch}(\mathrm{Sym}^2),$$

which proves (25), since finite-dimensional rational $\mathrm{GL}_2(\mathbb{C})$ -representations are semisimple and determined by their characters.

The first identity is tautological. To obtain the remaining identities, first note that

$$x^4(x^3y)(x^2y^2)(xy^3)y^4 = (xy)^{10},$$

so $\det(\mathrm{Sym}^4 S) = (\det S)^{10}$. For a rank-five bundle E and $0 \leq p \leq 5$, the canonical duality is

$$\bigwedge^{5-p} E \cong \bigwedge^p E^* \otimes \det E.$$

Applying this with $E = \mathrm{Sym}^4 S$ and $p = 2, 1$, respectively, and using $(\mathrm{Sym}^a S)^* \cong \mathrm{Sym}^a S \otimes (\det S)^{-a}$ for rank two, gives (26) and (27). Finally, $\bigwedge^5 \mathrm{Sym}^4 S = \det(\mathrm{Sym}^4 S)$ gives (28). Each identity, proved for the standard representation, holds for S by the associated-bundle construction. $\square$

For integers $\lambda_1 \geq \lambda_2$, including negative values, set

$$\mathbf{S}_{(\lambda_1, \lambda_2)} \mathcal{S}^* = \mathrm{Sym}^{\lambda_1 - \lambda_2} \mathcal{S}^* \otimes (\det \mathcal{S}^*)^{\lambda_2}. \quad (29)$$

For rank two, integers $a \geq 0$ and b satisfy the weight conversion

$$\mathrm{Sym}^a \mathcal{S} \otimes (\det \mathcal{S})^b \otimes \det \mathcal{S}^* \cong \mathbf{S}_{(1-b, 1-a-b)} \mathcal{S}^*, \quad (30)$$

because $\mathcal{S} \cong \mathcal{S}^* \otimes \det \mathcal{S}$. For example, the first summand of K_2 has $a = 6$ and $b = 1$, so

$$\mathrm{Sym}^6 \mathcal{S} \otimes \det \mathcal{S} \otimes \det \mathcal{S}^* \cong \mathbf{S}_{(0, -6)} \mathcal{S}^*.$$

Applying the same conversion to every summand in [Lemma 7.1](#) gives the following weights:

p	weights $\lambda = (\lambda_1, \lambda_2)$	
1	$(1, -3)$	
2	$(0, -6), (-2, -4)$	
3	$(-2, -8), (-4, -6)$	
4	$(-5, -9)$	
5	$(-9, -9)$	

(31)

7.2. The linear twist. We use the degree part of Borel–Weil–Bott in the convention fixed above: $G = \text{Gr}(2, V)$ parametrizes two-dimensional subspaces of V , and $\mathcal{S}$ is the tautological subbundle. Translating the homogeneous-bundle notation in Weyman’s formulation through [\(29\)](#) gives the following statement [[21](#), Corollary 4.1.9].

Theorem 7.2 (Borel–Weil–Bott on $\text{Gr}(2, N)$). *For $\lambda = (\lambda_1, \lambda_2)$ with $\lambda_1 \geq \lambda_2$, form*

$$A_N(\lambda) = (\lambda_1 + N - 1, \lambda_2 + N - 2, N - 3, \dots, 0). \quad (32)$$

If two entries coincide, all cohomology of $\mathbf{S}_\lambda \mathcal{S}^$ vanishes. Otherwise it is concentrated in the degree equal to the number of pairs $i < j$ for which the i th entry of $A_N(\lambda)$ is smaller than the j th entry; equivalently, this is the number of inversions needed to arrange the sequence in decreasing order.*

Only the vanishing and the unique possible cohomological degree are recorded here. Borel–Weil–Bott also identifies the resulting representation, but that identification is not needed below.

In the Bott tables below, *singular* means that $A_N(\lambda)$ has a repeated entry, so all cohomology groups of that summand vanish. The last column records the group $H^{p-1}(G, \mathbf{S}_\lambda \mathcal{S}^*)$ that must vanish for the summand in that row; here and below H^q suppresses the space and the bundle.

Proposition 7.3. *For every $N \geq 5$, the bundles K_p in [\(22\)](#) satisfy*

$$H^{p-1}(G, K_p) = 0 \quad (1 \leq p \leq 5).$$

Proof. We apply [Theorem 7.2](#) to the seven weights in [\(31\)](#); the case distinction is carried out in [Table 1](#).

p	λ	Borel–Weil–Bott outcome	needed group
1	$(1, -3)$	singular for every $N \geq 5$	$H^0 = 0$
2	$(0, -6)$	$q = N - 2$ for $5 \leq N \leq 7$; singular for $N \geq 8$	$H^1 = 0$
2	$(-2, -4)$	singular for every $N \geq 5$	$H^1 = 0$
3	$(-2, -8)$	singular for every $N \geq 5$	$H^2 = 0$
3	$(-4, -6)$	singular for every $N \geq 5$	$H^2 = 0$
4	$(-5, -9)$	$q = 6$ for $N = 5$; singular for $N \geq 6$	$H^3 = 0$
5	$(-9, -9)$	$q = 2N - 4$ for $5 \leq N \leq 9$; singular for $N \geq 10$	$H^4 = 0$

TABLE 1. The Borel–Weil–Bott outcomes for the summands of K_p .

The singular rows in [Table 1](#) come from repeated tail entries. For $\lambda = (1, -3)$ and $(-4, -6)$ the repeated entry is $N - 5$, while for $\lambda = (-2, -4)$ and $(-2, -8)$ it is $N - 3$. For the remaining three weights, the entries $N - 8$, $N - 6$, and $N - 10$ repeat when $N \geq 8$, $N \geq 6$, and $N \geq 10$, respectively.

It remains to count inversions in the nonsingular ranges. For $\lambda = (0, -6)$,

$$A_N(0, -6) = (N - 1, N - 8, N - 3, \dots, 0).$$

When $5 \leq N \leq 7$, the second entry lies below all $N - 2$ tail entries, so the cohomological degree is $q = N - 2$. For $\lambda = (-5, -9)$ and $N = 5$, both initial entries lie below the three tail entries, giving six inversions. For $\lambda = (-9, -9)$ and $5 \leq N \leq 9$, both initial entries lie below all $N - 2$ tail entries, giving $2N - 4$ inversions. None of these degrees is $p - 1$, and therefore every required group vanishes. $\square$

Proposition 7.4. *Let V be an N -dimensional vector space with $N \geq 5$, put $G = \text{Gr}(2, V)$, and let $Z \subset G$ be the zero scheme of a regular section of $\text{Sym}^4 \mathcal{S}^*$. Then*

$$H^0(G, \mathcal{I}_Z(1)) = 0. \quad (33)$$

If Z is reduced, then its closed points span the Plücker projective space $\mathbb{P}(\wedge^2 V)$.

Proof. In the resolution (21), let $C_1 = \mathcal{I}_Z(1)$, let $C_p = \text{im}(K_p \rightarrow K_{p-1})$ for $2 \leq p \leq 4$, and put $C_5 = K_5$. Exactness yields short exact sequences

$$0 \longrightarrow C_{p+1} \longrightarrow K_p \longrightarrow C_p \longrightarrow 0, \quad 1 \leq p \leq 4. \quad (34)$$

All cohomology groups in the following display are taken on G . Feeding Proposition 7.3 into the long exact cohomology sequences of (34), from $p = 4$ downwards, gives

$$\begin{aligned} H^3(C_4) &= 0, & \text{since } H^3(K_4) &= H^4(K_5) = 0; \\ H^2(C_3) &= 0, & \text{since } H^2(K_3) &= H^3(C_4) = 0; \\ H^1(C_2) &= 0, & \text{since } H^1(K_2) &= H^2(C_3) = 0; \\ H^0(C_1) &= 0, & \text{since } H^0(K_1) &= H^1(C_2) = 0. \end{aligned}$$

The last line is (33).

A Plücker hyperplane is the zero divisor of a nonzero section of $\mathcal{O}_G(1)$. If Z were reduced with all its closed points in such a hyperplane, the defining section would vanish on the reduced subscheme Z and would therefore lie in $H^0(G, \mathcal{I}_Z(1))$, contradicting (33). So no hyperplane contains the Plücker image of Z , and the points of Z span $\mathbb{P}(\wedge^2 V)$. $\square$

7.3. The quadratic twist on $\text{Gr}(2, 5)$.

Proposition 7.5. *Let $G = \text{Gr}(2, 5)$, and let $Z \subset G$ be the zero scheme of a regular section of $\text{Sym}^4 \mathcal{S}^*$. Then*

$$H^0(G, \mathcal{I}_Z(2)) = 0. \quad (35)$$

In particular, this applies to $Z = F_1(X)$ for every smooth quartic threefold $X \subset \mathbb{P}^4$.

Proof. Twisting the Koszul resolution of the regular section by $\mathcal{O}_G(2) = (\det \mathcal{S}^*)^{\otimes 2}$ gives an exact sequence

$$0 \longrightarrow K_5^{(2)} \longrightarrow K_4^{(2)} \longrightarrow K_3^{(2)} \longrightarrow K_2^{(2)} \longrightarrow K_1^{(2)} \longrightarrow \mathcal{I}_Z(2) \longrightarrow 0, \quad (36)$$

where

$$K_p^{(2)} = \bigwedge^p (\text{Sym}^4 \mathcal{S}) \otimes (\det \mathcal{S}^*)^{\otimes 2}.$$

Relative to the weights in (31), the extra factor $\det \mathcal{S}^*$ adds $(1, 1)$; Theorem 7.2 with $N = 5$ gives the following list.

The repeated entries displayed in Table 2 give the singular rows, and the three nonsingular rows have three, three, and six inversions respectively. So

$$H^{p-1}(G, K_p^{(2)}) = 0 \quad (1 \leq p \leq 5). \quad (37)$$

p	λ	$A_5(\lambda)$	Borel–Weil–Bott outcome	needed group
1	(2, -2)	(6, 1, 2, 1, 0)	singular	$H^0 = 0$
2	(1, -5)	(5, -2, 2, 1, 0)	$q = 3$	$H^1 = 0$
2	(-1, -3)	(3, 0, 2, 1, 0)	singular	$H^1 = 0$
3	(-1, -7)	(3, -4, 2, 1, 0)	$q = 3$	$H^2 = 0$
3	(-3, -5)	(1, -2, 2, 1, 0)	singular	$H^2 = 0$
4	(-4, -8)	(0, -5, 2, 1, 0)	singular	$H^3 = 0$
5	(-8, -8)	(-4, -5, 2, 1, 0)	$q = 6$	$H^4 = 0$

TABLE 2. The quadratic-twist calculation on $\text{Gr}(2, 5)$.

The dimension-shifting argument of the proof of [Proposition 7.4](#), applied to (36) and (37), now gives $H^0(G, \mathcal{I}_Z(2)) = 0$. The final assertion is [Proposition 6.1](#). $\square$

8. ACTUAL PLÜCKER SPANNING

For $n \geq 5$, [Propositions 6.1](#) and [7.4](#) already give spanning by actual lines. It remains to treat a smooth quartic threefold X , whose Fano scheme $Z = F_1(X)$ may be nonreduced. We bound the generic lengths of its components, deduce that its nilradical has square zero, and apply [Proposition 7.5](#).

Call an irreducible component of Z *exotic* if its general line has the completely unbalanced normal bundle $\mathcal{O}_L(1) \oplus \mathcal{O}_L(-2)$ of (18), which for $n = 4$ is the second type in (20). For the classical description of exotic components, see [16, § 1, (1.2)(3)]; the local dichotomy needed here is derived in the proof below. For a curve $T \subset Z_{\text{red}}$, its *swept surface* is the reduced Zariski closure of the union of the lines it parametrizes. A *cone component* is one whose general lines pass through a fixed point, called its vertex. The *tangent developable* of a curve is the reduced closure of the union of its tangent lines at smooth points. For a component with generic point η , its *generic length* means $\text{length}_{\mathcal{O}_{Z,\eta}} \mathcal{O}_{Z,\eta}$.

Lemma 8.1. *Let $X \subset \mathbb{P}^4$ be a smooth quartic threefold, put $Z = F_1(X)$, and let η_T be the generic point of an irreducible component $T \subset Z_{\text{red}}$. Then*

$$\text{length}_{\mathcal{O}_{Z,\eta_T}} \in \{1, 2\}.$$

The length is one if the general line of T has normal bundle $\mathcal{O}_L \oplus \mathcal{O}_L(-1)$, and two if it has normal bundle $\mathcal{O}_L(1) \oplus \mathcal{O}_L(-2)$.

Proof. By [Proposition 6.1](#), Z is a pure curve, and the two normal-bundle types in (20) exhaust the possibilities for a line on X . If $N_{L/X} \cong \mathcal{O}_L \oplus \mathcal{O}_L(-1)$ for a general line of T , then $\dim T_{[L]}Z = h^0(L, N_{L/X}) = 1$, so Z is smooth at a general point of T and its generic length is one. We may therefore assume that T is exotic. Write $X = \{F = 0\} \subset \mathbb{P}(V)$ with $\dim V = 5$.

We will use that X contains no plane. The two cubic coefficients of F along such a plane would have a common zero, making X singular; see [2, proof of Theorem 4.2, case $n = d = 4$].

Step 1: local forms of the actual family. Work on a smooth analytic parameter disc in T , away from its intersections with the other components of Z_{red} , on which the normal bundle has the exotic type, and write its lines as $L_t = \mathbb{P}(U_t)$. Denote the disc by Δ and the corresponding incidence surface by

$$q : \mathcal{L} = \{(x, t) \in X \times \Delta : x \in L_t\} \longrightarrow \Delta.$$

This is a holomorphic $\mathbb{P}^1$ -bundle. The normal bundles of its fibres form a rank-two holomorphic bundle

$$\mathcal{N} = \text{pr}_X^* TX / T_{\mathcal{L}/\Delta} \quad \text{on } \mathcal{L},$$

with $\mathcal{N}|_{L_t} = N_{L_t/X}$. In a local trivialization of q , differentiating the evaluation map $\mathcal{L} \rightarrow X$ in the t -direction gives a holomorphic section s of $\mathcal{N}$; changing the trivialization adds a relative tangent vector and hence does not change its class. The restriction of s is the nonzero characteristic section

$$s_t \in H^0(L_t, N_{L_t/X}).$$

We next show that the zero of s_t varies holomorphically. Since $N_{L_t/X} \cong \mathcal{O}_{L_t}(1) \oplus \mathcal{O}_{L_t}(-2)$ throughout Δ , the cohomology and base change theorem makes $q_*\mathcal{N}$ a rank-two bundle with fibre $H^0(L_t, N_{L_t/X})$. The evaluation morphism

$$q^*q_*\mathcal{N} \longrightarrow \mathcal{N}$$

has constant rank one: on every fibre its image is precisely the $\mathcal{O}_{L_t}(1)$ summand. Its image is therefore a holomorphic line subbundle $\mathcal{M} \subset \mathcal{N}$, with $\mathcal{M}|_{L_t} \cong \mathcal{O}_{L_t}(1)$, and the section s takes values in $\mathcal{M}$. Consequently s_t has exactly one simple zero on every fibre. After shrinking Δ , the holomorphic implicit-function theorem shows that the zero locus of s is the graph of a holomorphic section $z(t) \in L_t$. Let $p(t) \in U_t \setminus \{0\}$ be a holomorphic vector lift of $z(t)$. Then $s_t(z(t))$ is identified with $p'(t) \bmod U_t$, and $p'(t) \in U_t$ because $z(t)$ is a zero of s_t . If $z(t)$ is fixed, choose holomorphic lifts so that

$$U_t = \text{Span}(x, c(t)), \quad x \in V \setminus \{0\} \text{ fixed.}$$

Otherwise $p(t)$ and $p'(t)$ are independent at a general parameter, and

$$U_t = \text{Span}(p(t), p'(t)).$$

These are the cone and tangent-developable cases, respectively.

On the disc, T maps into the Grassmannian with nonzero differential. Hence x, c, c' are independent in the cone case and p, p', p'' are independent in the tangent case. At a general parameter, more is true:

$$x, c, c', c'' \quad \text{or} \quad p, p', p'', p''' \quad \text{are independent at a general parameter.} \quad (38)$$

Consider the cone case. The derivative of $t \mapsto \text{Span}(x, c, c') \in \text{Gr}(3, V)$ is represented by c'' modulo $\text{Span}(x, c, c')$, so if c'' belonged to this subspace at a general parameter, the map would be constant. The lines L_t would then sweep out its fixed projective plane, and that plane would lie in X . In the tangent-developable case the same argument applies to $t \mapsto \text{Span}(p, p', p'')$, whose derivative is represented by p''' modulo that subspace.

Step 2: gradient identities along a general line. In the cone case put $r_t(u, v) = ux + vc(t)$, where $[u : v]$ are coordinates on L_t . Differentiating $F(r_t(u, v)) = 0$ in t and cancelling the polynomial factor v gives

$$dF_{r_t}(c') = 0, \quad dF_{r_t}(c'') = -vD^2F_{r_t}(c', c').$$

Here D^2F is the Hessian of F , viewed as a symmetric bilinear form. In the tangent-developable case put $r_t(u, v) = up(t) + vp'(t)$. Since $F|_{U_t} = 0$, the differential dF_{r_t} annihilates p and p' , and differentiating $F(r_t) = 0$ in t gives $0 = v dF_{r_t}(p'')$. Cancelling the polynomial factor v yields $dF_{r_t}(p'') = 0$, and differentiating this identity in v gives

$$D^2F_{r_t}(p', p'') = 0.$$

Differentiating $dF_{r_t}(p'') = 0$ in t therefore gives

$$dF_{r_t}(p''') = -uD^2F_{r_t}(p', p'') - vD^2F_{r_t}(p'', p'') = -vD^2F_{r_t}(p'', p'').$$

Fix a general parameter, call it zero, and put $L = \mathbb{P}(U_0)$ and $r = r_0$. In the cone case set $e_0 = c'(0)$ and $e_1 = c''(0)$; in the tangent-developable case set $e_0 = p''(0)$ and $e_1 = p'''(0)$. By (38) we can add e_2 to obtain a fixed complement $\text{Span}(e_0, e_1, e_2)$ of U_0 in V . Define the binary forms

$$B_2(u, v) = D^2 F_r(e_0, e_0), \quad C_3(u, v) = dF_r(e_2),$$

of degrees two and three. The identities above become

$$dF_r(e_0) = 0, \quad dF_r(e_1) = -vB_2, \quad dF_r(e_2) = C_3. \quad (39)$$

Since dF_r also annihilates U_0 , a common projective zero of vB_2 and C_3 would be a singular point of X . Smoothness along the entire line therefore gives

$$\gcd(vB_2, C_3) = 1, \quad B_2 \neq 0, \quad C_3 \neq 0. \quad (40)$$

If either form were zero, the other cubic would have a projective zero. In particular $\gcd(B_2, C_3) = 1$ and $C_3(1, 0) \neq 0$.

Step 3: elimination and the surviving quadratic term. Let $R_j = \mathbb{C}[u, v]_j$ be the vector space of homogeneous polynomials of degree j . In the Grassmannian chart defined by the chosen complement, nearby lines are the graphs

$$r(u, v) + L_0(u, v)e_0 + L_1(u, v)e_1 + L_2(u, v)e_2, \quad L_i \in R_1.$$

Substitution in F gives a polynomial map $\mathcal{E} : R_1^3 \rightarrow R_4$ whose five coefficients cut out Z in this chart. By (39) its differential at the origin is

$$d\mathcal{E}_0(L_0, L_1, L_2) = -vB_2L_1 + C_3L_2.$$

By (40), the restriction of $d\mathcal{E}_0$ to the four coefficients of (L_1, L_2) is injective, so its kernel is exactly the two L_0 -directions and its image is the four-dimensional subspace

$$\mathcal{L} = vB_2R_1 + C_3R_1 \subset R_4.$$

Moreover,

$$u^2B_2, \quad uvB_2, \quad v^2B_2, \quad uC_3, \quad vC_3$$

form a basis of R_4 : a dependence would give $Q_2B_2 + Q_1C_3 = 0$ with $Q_i \in R_i$, and $\gcd(B_2, C_3) = 1$ would force the cubic C_3 to divide the quadratic Q_2 , hence $Q_2 = Q_1 = 0$.

Let $\lambda : R_4 \rightarrow \mathbb{C}$ be the linear functional that vanishes on $\mathcal{L}$ and satisfies $\lambda(u^2B_2) = 1$. Then

$$P(q) = q - \lambda(q)u^2B_2$$

is the projection from R_4 to $\mathcal{L}$. The four components of $P\mathcal{E}$ have invertible Jacobian in the four coefficients of (L_1, L_2) and cut out a smooth algebraic surface Σ near $[L]$. Write $L_0 = au + bv$ and let z denote those four coefficients. In the completed local ring of Σ the formal implicit-function theorem gives $z = \zeta(a, b)$ and $\widehat{\mathcal{O}}_{\Sigma, [L]} \cong \mathbb{C}[[a, b]]$. Since $d\mathcal{E}_0$ vanishes in the L_0 -directions, $\zeta(0) = 0$ and $D\zeta(0) = 0$.

On Σ the remaining equation is the algebraic function $f = (\lambda\mathcal{E})|_{\Sigma}$, which in the completed local ring gives

$$\widehat{\mathcal{O}}_{\Sigma, [L]} \cong \mathbb{C}[[a, b]]/(\varphi), \quad \varphi(a, b) = \lambda(\mathcal{E}(a, b, \zeta(a, b))). \quad (41)$$

Here the completion is at the maximal ideal corresponding to $[L]$. The series φ has no linear term, since $\lambda d\mathcal{E}_0 = 0$, and the corrections from ζ do not affect its quadratic part: ζ has order at least two, so its contribution through $d\mathcal{E}_0$ lies in $\mathcal{L}$ and is killed by λ , terms mixing L_0 with ζ have order at least three, and terms quadratic in ζ have order at least four. The basis above gives $\lambda(uvB_2) = \lambda(v^2B_2) = 0$ and $\lambda(u^2B_2) = 1$, so the quadratic part of φ is

$$\varphi_2(a, b) = \frac{1}{2}\lambda((au + bv)^2B_2) = \frac{1}{2}a^2. \quad (42)$$

In particular, the transverse corrections made in eliminating the four variables do not change this nonzero quadratic term.

Step 4: from the quadratic term to generic length. Set $A_\Sigma = \mathcal{O}_{\Sigma, [L]}$, a regular local ring of dimension two, and let $\mathfrak{p}$ be the ideal of T in this local ring. At the chosen point $[L]$ the reduced support of the hypersurface $V(f) \subset \Sigma$ is exactly the smooth curve T , because no other component of Z_{red} passes through $[L]$. Therefore

$$\sqrt{(f)} = \mathfrak{p}, \quad \text{height } \mathfrak{p} = 1.$$

A regular local ring is factorial [20, Lemma 15.123.2]. Hence

$$\mathfrak{p} = (h), \quad f = \epsilon h^m \quad \text{for some } \epsilon \in A_\Sigma^\times, \quad m \geq 1.$$

Since T is a smooth divisor in the smooth surface Σ at $[L]$, its local equation satisfies $h \in \mathfrak{m}_{A_\Sigma} \setminus \mathfrak{m}_{A_\Sigma}^2$. Thus h has maximal-ideal order one and $f = \epsilon h^m$ has order m . Completion is faithfully flat and therefore preserves this order, while (42) shows that the completed equation has order exactly two. Hence $m = 2$. Localizing at $\mathfrak{p}$ now gives

$$\text{length } \mathcal{O}_{Z, \eta_T} = \text{length}_{(A_\Sigma)_\mathfrak{p}}(A_\Sigma)_\mathfrak{p}/(f) = m = 2,$$

because $(A_\Sigma)_\mathfrak{p}$ is a discrete valuation ring with uniformizer h . This proves the assertion for every exotic component. $\square$

Corollary 8.2 (Square-zero nilradical). *Let $X \subset \mathbb{P}^4$ be a smooth quartic threefold and put $Z = F_1(X)$. The nilradical*

$$\mathcal{N} = \ker(\mathcal{O}_Z \longrightarrow \mathcal{O}_{Z_{\text{red}}})$$

satisfies $\mathcal{N}^2 = 0$.

Proof. By Lemma 8.1, each generic local ring of Z is Artinian of length one or two, with nilradical its maximal ideal $\mathfrak{m}$. In length one, $\mathfrak{m} = 0$; in length two, $\mathfrak{m}$ has length one, so $\mathfrak{m}^2 \neq 0$ would force $\mathfrak{m}^2 = \mathfrak{m}$, contradicting Nakayama's lemma. Thus $\mathcal{N}^2$ vanishes at every generic point.

By Proposition 6.1, Z is a pure one-dimensional Cohen–Macaulay scheme, so its associated points are exactly its component generic points. A nonzero coherent ideal subsheaf $\mathcal{N}^2 \subset \mathcal{O}_Z$ would have an associated point, necessarily an associated point of $\mathcal{O}_Z$, contradicting its vanishing at all those points. Hence $\mathcal{N}^2 = 0$ on Z . $\square$

Theorem 8.3. *Let V be a complex vector space of dimension $n+1$ with $n \geq 4$, and let $X \subset \mathbb{P}(V)$ be a smooth quartic hypersurface. Then the bivectors $a \wedge b$ of the actual lines $\mathbb{P}(\text{Span}(a, b)) \subset X$ span $\bigwedge^2 V$:*

$$\text{Span}\{a \wedge b : \mathbb{P}(\text{Span}(a, b)) \subset X, \ a, b \text{ linearly independent}\} = \bigwedge^2 V.$$

In particular one can choose $\binom{n+1}{2}$ distinct actual lines on X whose Plücker bivectors form a basis.

Proof. Put $Z = F_1(X) \subset G = \text{Gr}(2, V)$. For $n \geq 5$, Proposition 6.1 makes Z a reduced regular zero locus and Proposition 7.4 gives the asserted spanning.

Suppose that $n = 4$. Under the Plücker identification $H^0(G, \mathcal{O}_G(1)) \cong \bigwedge^2 V^*$, if the actual line bivectors failed to span $\bigwedge^2 V$, there would be a nonzero section $h \in H^0(G, \mathcal{O}_G(1))$ vanishing on Z_{red} . Its restriction to Z lies in $\mathcal{N} \otimes \mathcal{O}_Z(1)$, because $\mathcal{N}$ is the ideal of Z_{red} in Z . By Corollary 8.2 the section $h^2|_Z$ lies in $\mathcal{N}^2 \otimes \mathcal{O}_Z(2) = 0$; since the Grassmannian is integral, $h^2 \neq 0$ on G , so h^2 is a nonzero element of $H^0(G, \mathcal{I}_Z(2))$, contrary to Proposition 7.5. This proves spanning in all cases. Extracting a basis from the actual line bivectors gives the stated number of distinct lines. $\square$

9. PROOF OF THE MAIN THEOREM AND FURTHER REMARKS

Proof of Theorem 1.1. Write $X \subset \mathbb{P}(V)$ with $\dim V = n + 1$. By Theorem 8.3 we may choose $\binom{n+1}{2}$ actual lines whose Plücker bivectors form a basis of $\bigwedge^2 V$. Their line-projection sprays are defined on $\mathcal{O}_X(-2)$ by Proposition 4.4 and form a dominating family by Proposition 5.3, so X is holomorphically subelliptic. The Oka property and holomorphic ellipticity now follow from Theorems 2.3 and 2.4. $\square$

Remark 9.1 (Why degree four is special). For a smooth degree- d hypersurface $X_d \subset \mathbb{P}^n$, $n \geq 4$ and $d \geq 2$, containing a line, the same calculation gives

$$K_Y^{-1} \otimes \pi^* K_{\mathbb{P}^{n-2}} \cong \mathcal{O}_Y((4-d)H) \otimes \pi^* \mathcal{O}_{\mathbb{P}^{n-2}}(-2).$$

Its restriction to a general smooth residual plane curve of degree $d-1$ has degree $(4-d)(d-1)$, so the time bundle is pulled back from the base exactly when $d=4$. For $d \geq 5$ the residual curve has genus greater than one and carries no nonzero holomorphic vector field. This concerns the present construction only, not other approaches to the Oka property.

Remark 9.2 (Quartic surfaces). The uniform Plücker-spanning theorem above applies when $n \geq 4$; a general quartic surface in $\mathbb{P}^3$ contains no line. If a smooth quartic surface $S \subset \mathbb{P}^3$ contains a line L , cancelling the common section σ_L from the two linear forms generating $\mathcal{I}_{L/S}(1)$ gives a base-point-free pencil $\pi_L : S \rightarrow \mathbb{P}^1$ with $\pi_L^* \mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{O}_S(H_S - L)$, where H_S is the hyperplane class. Since $K_S \cong \mathcal{O}_S$, the determinant source is $\pi_L^* K_{\mathbb{P}^1} \cong \mathcal{O}_S(-2H_S + 2L)$. Multiplication by σ_L^2 maps $\mathcal{O}_S(-2H_S)$ into this source and vanishes along L , so Lemma 3.1 and Proposition 3.2 give a spray on $\mathcal{O}_S(-2)$ that is constant over L .

Remark 9.3 (Nonalgebraicity of the constructed sprays). Fix a line $L \subset X$ and a general smooth residual cubic C_b as in Lemma 4.2. For $x \in C_b \setminus L$, trivialize $\mathcal{O}_X(-2)_x$ and consider the trajectory $t \mapsto s_L(x, t)$. It lies in C_b and has nonzero derivative at zero by Proposition 4.4. The trajectory cannot be algebraic: an algebraic map $\mathbb{A}^1 \rightarrow C_b$ extends to $\mathbb{P}^1 \rightarrow C_b$, which is constant by Riemann–Hurwitz. Thus the sprays are holomorphic and not algebraic. This does not decide whether X has a different algebraic dominating spray.

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DECLARATION ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE

Generative artificial intelligence assisted with drafting and language editing. The mathematical ideas, proof strategy, and conclusions are those of the authors, who reviewed and verified all AI-assisted material and take full responsibility for the manuscript.

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